

RAG

Retrieval-Augmented Generation

What is RAG?

Retrieval-Augmented Generation (**RAG**) is a framework designed to give Large Language Models (**LLMs**) access to data they weren't originally trained on

For example:

- LLM wasn't trained to know an information related to company, it could be some reports or confidential info and this data should have restricted access.
- You created a big documentation and want open LLM access to it
- Latest news, Stocks...



Vector Embedding

Text to Tokens to Vectors



Model converts text to tokens, then maps them to specific vectors.

High-Dimensional Space



Maps tokens to points in a high-dimensional space.

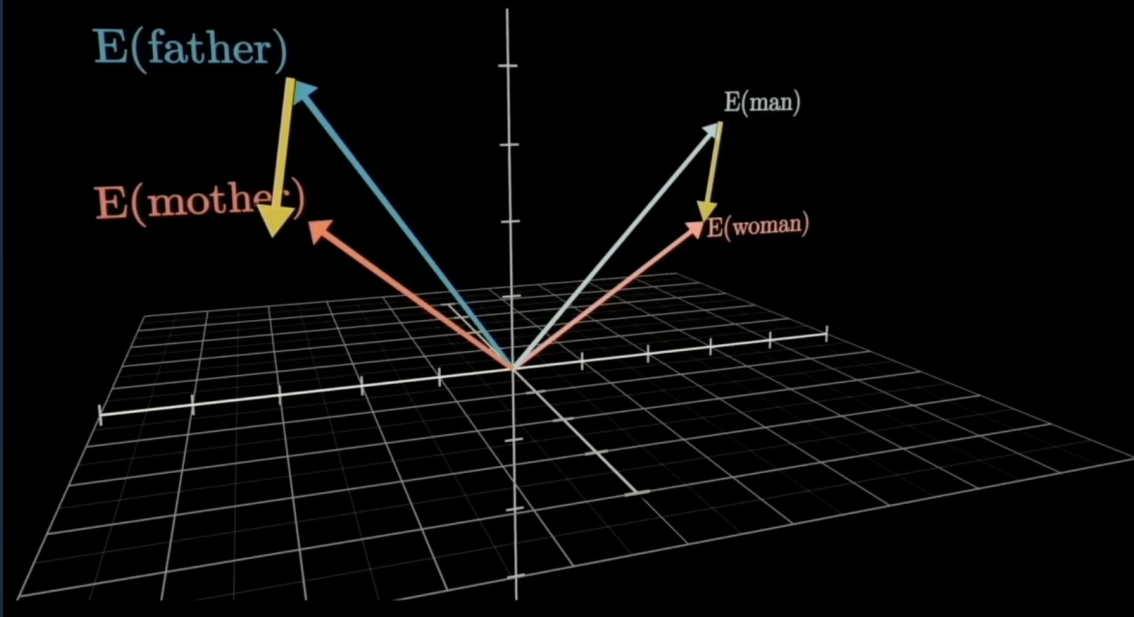
Semantic Meaning



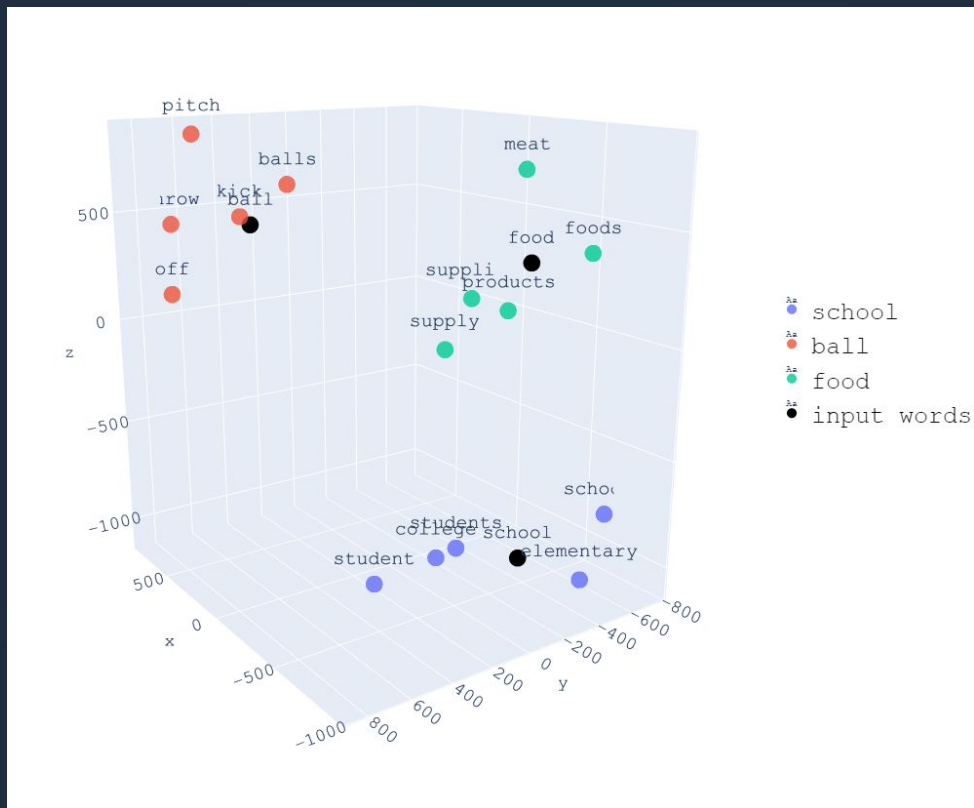
Mathematically close vectors for similar concepts.

Vector Embedding

$$E(\text{mother}) - E(\text{father}) \approx E(\text{woman}) - E(\text{man})$$



Vector DB



Chunking

Chunk is a smaller, segment of a larger document, created by a process called chunking, to provide relevant context to a LLM for more accurate and efficient responses.



Chunking

Chunk is a smaller, segment of a larger chunking, to provide relevant context to responses.

Code Example:

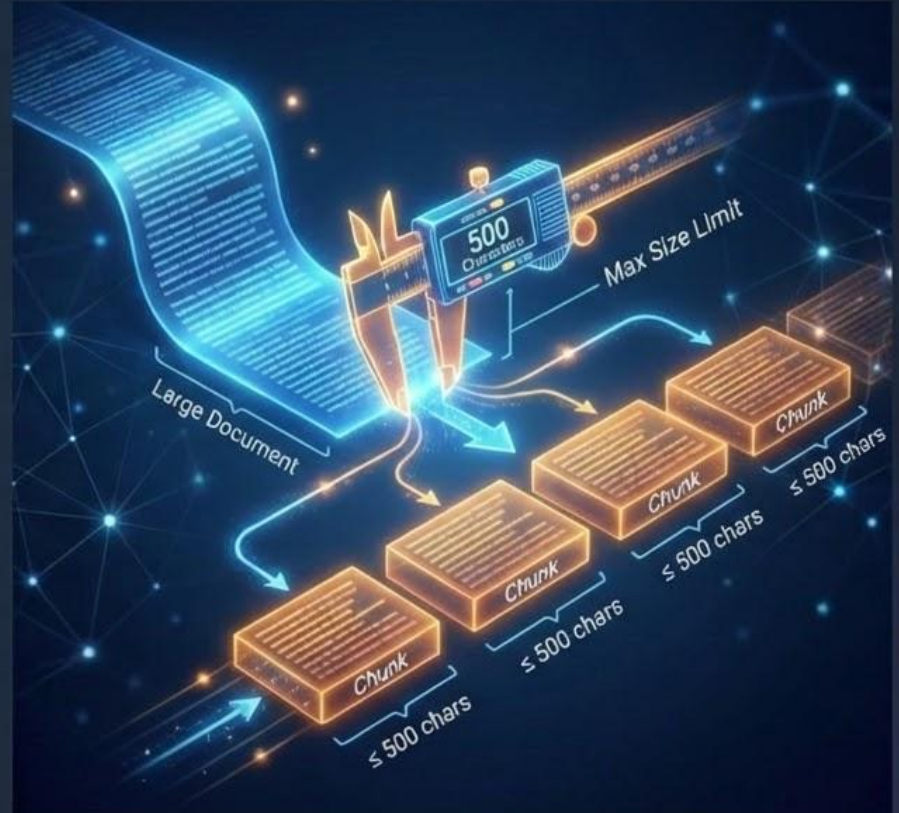
```
RecursiveCharacterTextSplitter(  
    chunk_size=500,  
    chunk_overlap=50,  
    separators=["\n\n", "\n", ". ", " "]  
)
```



Chunking

`chunk_size=500`

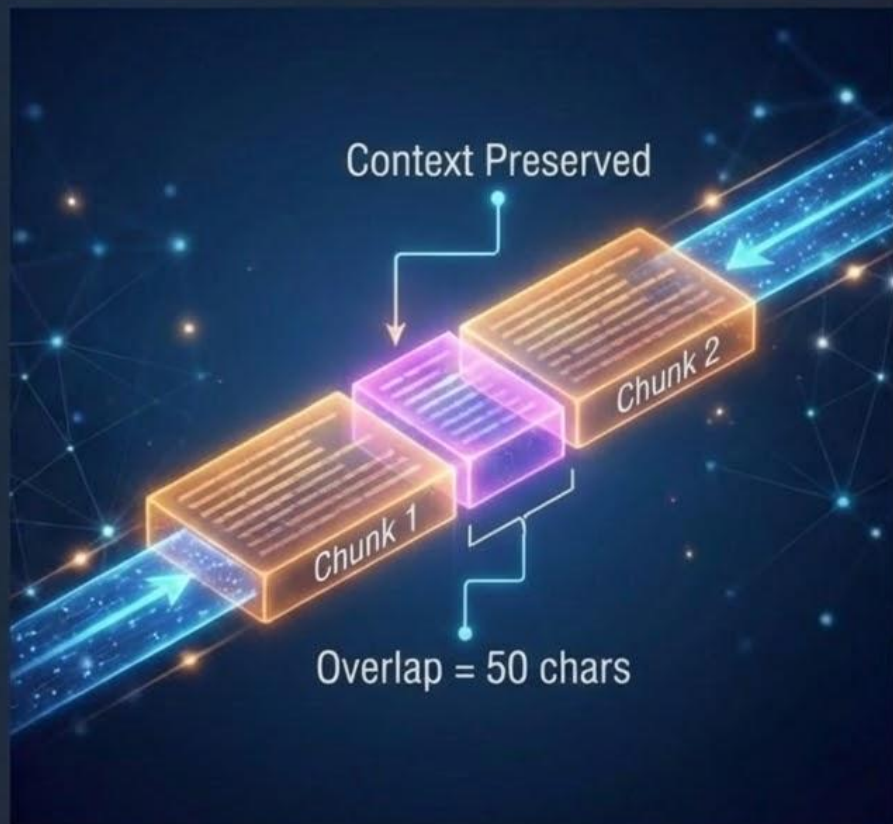
Maximum characters per chunk. Each chunk will be at most 500 characters.



Chunking

chunk_overlap=50

Characters that overlap between consecutive chunks. This ensures context isn't lost at boundaries.



Chunking

For example: chunk_size=20 and overlap=5

Text: "AAAAAAAAAA BBBBBBBBBB CCCCCCCCCC DDDDDDDDDD"

Chunk 1: "AAAAAAAAAA BBBBBBBBBB"

Last 5 chars: "BBBBB" (Overlap)

5 copied to start

"BBBBB CCCCCCCCCC DDD" Chunk 2

Overlaps 5 characters

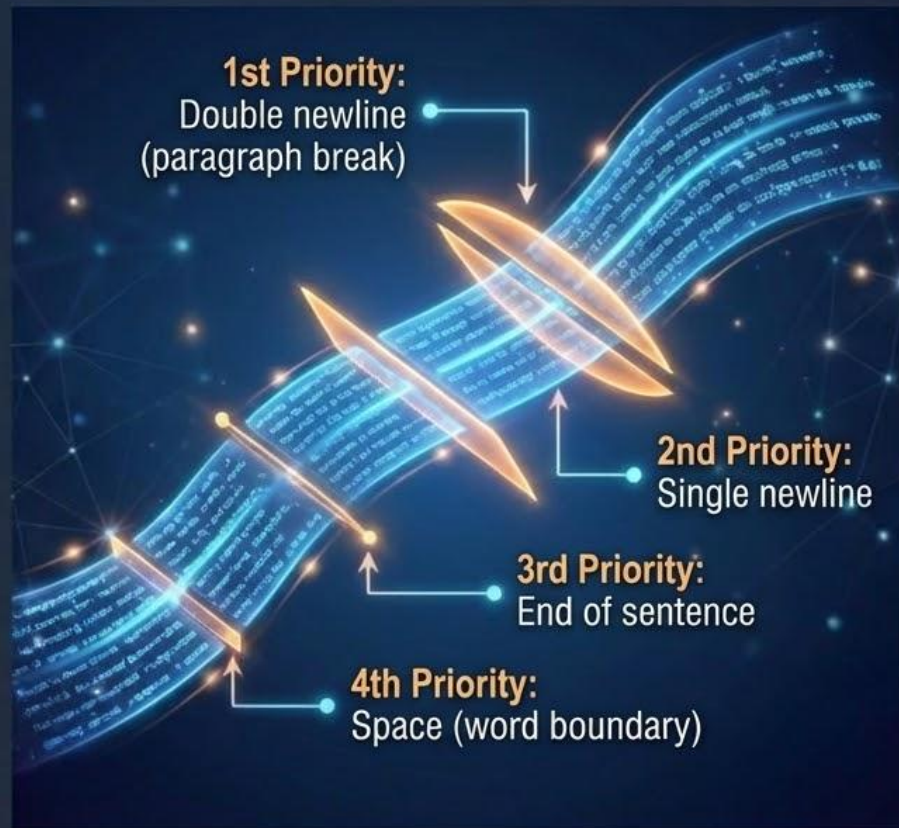
The overlap creates a "sliding window" where the tail of each chunk becomes the head of the next one. This ensures no context is lost at the boundaries!

Chunking

`separators=["\\n\\n", "\\n", " . ", " "]`

The splitter tries to split at these boundaries in order of priority:

1st	Double newline (\\n\\n)
2nd	Single newline (\\n)
3rd	End of sentence (.)
4th	Space ()



Chunking



“\n\n”

Paragraphs

Splits at: Paragraphs
Preserves: Complete ideas
Used when: First choice



“\n”

Lines

Splits at: Lines
Preserves: Related sentences
Used when: Paragraphs too big



“ ”

Sentences

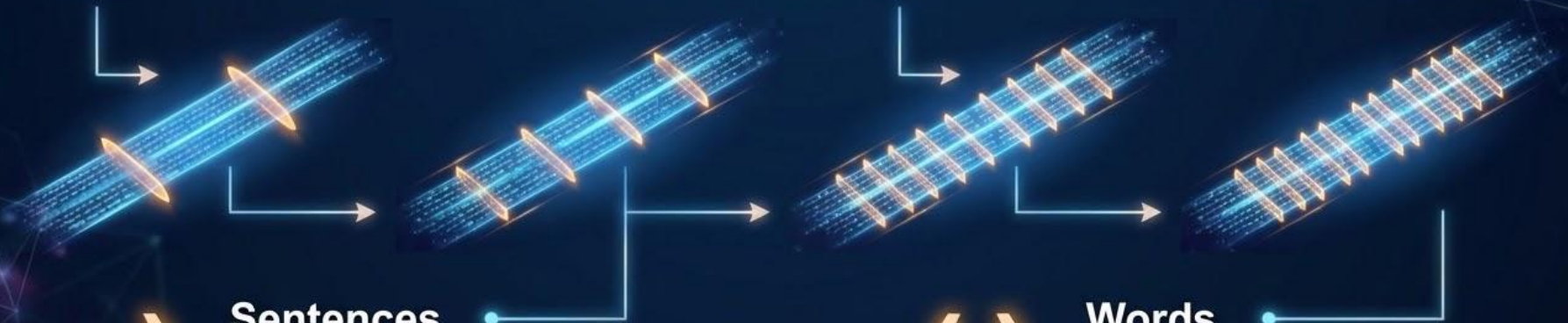
Splits at: Sentences
Preserves: Complete thoughts
Used when: Lines too big



“ ”

Words

Splits at: Words
Preserves: Whole words
Used when: Sentences too big



Vector DB

Helps to find a needed info/chunk using mathematics.

- Most popular algorithm is **HNSW (Hierarchical Navigable Small World)**
- HNSW Searching for the **closest by distance Vector**, going **layer by layer** (neighbors).



Vector DB



Imagine your database has **10 million vectors**. Without an index, the computer has to do 10 million math calculations for every search.

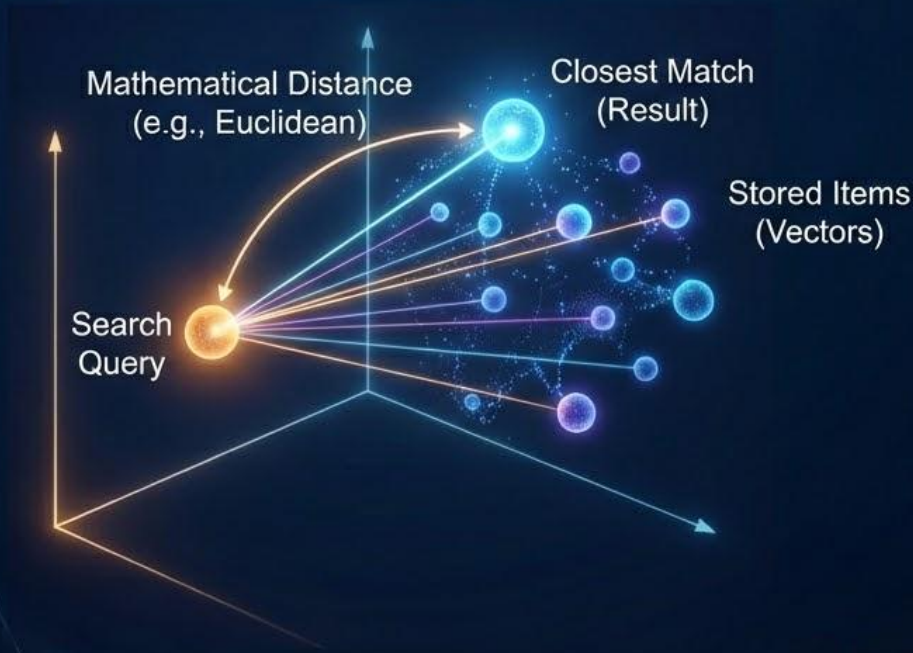
[HNSW wiki](#)



HNSW helps to organize data into specialized index structures to achieve **high speed data search**.

Response

A Vector Database finds results by **comparing the mathematical distance** between your search query and the items it has stored



Basic RAG Pipeline

